ML Cheat Sheet, Ch. 1–5

$x \in \mathbb{R}^d$ input, y label, $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$ i.i.d. $\sim \mathbb{P}$, $f_\theta \in \mathcal{H}$ model, $\hat{p} = \sigma(s)$ posterior, ℓ loss, R population risk, $\widehat{R}$ empirical risk, C classes.

1 Information

$$H(p) = -\sum_c p_c \log p_c$$ entropy

$$H(p,q) = -\sum_c p_c \log q_c = H(p) + \mathrm{KL}(p\|q)$$ cross-entropy

$$\mathrm{KL}(p\|q) = \sum_c p_c \log \frac{p_c}{q_c} \geq 0, \ = 0 \iff q = p$$ Gibbs ($\ln u \leq u - 1$)

2 Decisions: the loss decides

$$R(a \mid x) = \sum_y \ell(y,a)\, p(y \mid x)$$ conditional risk

$$f^*(x) = \arg\min_a R(a \mid x)$$ Bayes rule; pointwise $\Rightarrow$ global

$$0\text{–}1\text{:}\ f^* = \arg\max_c p(c \mid x)$$ error

$\mathbb{E}[1 - \max_c p(c \mid X)]$ = Bayes error

alert $\iff C_{FP}(1-p) < C_{FN}\, p$

$$\iff p > t = \frac{C_{FP}}{C_{FP} + C_{FN}}$$ cost threshold

Reject / review at fixed cost C_R

$$\text{benign if } p < \frac{C_R}{C_{FN}}, \quad \text{attack if } p > 1 - \frac{C_R}{C_{FP}}$$

review between

band exists iff $C_R < \frac{C_{FP}C_{FN}}{C_{FP}+C_{FN}}$ (the peak of $\min(C_{FN}p,\, C_{FP}(1-p))$)

Point prediction

$$\mathbb{E}[(Y-a)^2 \mid x] = \mathrm{Var}(Y \mid x) + (m(x) - a)^2$$

squared $\to$ mean $m(x)$

absolute $\to$ median; cost c_u per unit under, c_o over $\to$ quantile $\tau = \frac{c_u}{c_u + c_o}$, $F(a^*) = \tau$

k nearest neighbours

$$\hat{f}_k(x) = \frac{1}{k} \sum_{i \in N_k(x)} y^{(i)}$$ regression

$$\hat{p}_k(c \mid x) = \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}[y^{(i)} = c]$$ vote

$$d_p = \Big(\sum_j |x_j - z_j|^p \Big)^{1/p}$$ Minkowski, p=2 Euclid

$$d_w^2 = \sum_j w_j \Big(\frac{x_j - z_j}{s_j} \Big)^2$$ scale first; w chosen on validation

1-NN boundary: perpendicular bisector; capacity $\uparrow$ as $k \downarrow$ ($\approx N/k$ regions)

$M_d \sim \mathrm{Bin}(n, r^d)$, $\mathbb{P}(M_d \geq 1) = 1 - (1 - r^d)^n$ neighbours within r of the centre of the unit d-ball

$$r_d = \left(1 - 0.05^{1/n}\right)^{1/d} \to 1$$ radius for $\mathbb{P} = 0.95$: locality dies

2 Empirical risk, likelihood, evaluation

$$R(\theta) = \mathbb{E}\, \ell(Y, f_\theta(X))$$ population risk, unknowable

$$\widehat{R}(\theta) = \frac{1}{N} \sum_i \ell\big(y^{(i)}, f_\theta(x^{(i)})\big)$$ empirical risk

$$\hat{\theta} = \arg\min_\theta \widehat{R}(\theta)$$ ERM

$$\mathbb{E}[\widehat{R}(\hat{\theta})] \leq \min_\theta R(\theta) \leq \mathbb{E}[R(\hat{\theta})]$$ training error is optimistic

$$w = \frac{N\sum xy - \sum x \sum y}{N\sum x^2 - (\sum x)^2}, \quad b = \frac{\sum y - w \sum x}{N}$$ 1-D least squares

$$\hat{\theta} = (X^\top X)^{-1} X^\top y$$ normal equations

$$\hat{\theta}_{\text{ridge}} = (X^\top X + N\lambda I)^{-1} X^\top y$$

$\ell_{CE} = -\log \hat{p}_y(x)$ CE; $\frac{1}{N}\sum = -\frac{1}{N}$ log cond. likelihood: CE = MLE

$$\mathbb{E}[-\log \hat{p}_Y(x) \mid x] = H(p(x)) + \mathrm{KL}(p(x)\|\hat{p}(x))$$ min at the true posterior: calibration

$$\widehat{R}_{01} = \frac{1}{N} \sum_i \mathbf{1}[y^{(i)} \neq \hat{y}^{(i)}]$$ flat, cannot be optimised; CE and 0–1 can rank models differently

$$\hat{\theta}_\lambda = \arg\min_\theta \widehat{R}_{tr}(\theta; \lambda)$$ parameters on train

$$\hat{\lambda} = \arg\min_\lambda \widehat{R}_{val}(\hat{\theta}_\lambda)$$ hyperparameters on validation; test once

$$\widehat{R}_{CV} = \frac{1}{K} \sum_{j=1}^K \widehat{R}_{val}^{(j)}$$ fit scaler, imputer, feature selection inside each fold

$$z_j = \frac{x_j - \hat{\mu}_j}{\hat{\sigma}_j}$$ $\hat{\mu}, \hat{\sigma}$ from training rows only

Splits: random (exchangeable rows) / grouped (episodes, patients) / temporal (future $\neq$ past).

3 Generalization

$G = R(\hat{f}_\mathcal{D}) - \widehat{R}_\mathcal{D}(\hat{f}_\mathcal{D})$ gap; only $\widehat{R}$ is observable

$$\mathcal{H}_1 \subset \mathcal{H}_2 \Rightarrow \min_{\mathcal{H}_2} \widehat{R} \leq \min_{\mathcal{H}_1} \widehat{R}$$ train error never rises with capacity

Bias–variance at a fixed x, $Y = f(x) + \varepsilon$

$$\bar{f}(x) = \mathbb{E}_\mathcal{D} \hat{f}_\mathcal{D}(x), \quad \mathrm{Bias}(x) = \bar{f}(x) - f(x)$$

$$\mathrm{Var}(x) = \mathbb{E}_\mathcal{D}\big(\hat{f}_\mathcal{D}(x) - \bar{f}(x)\big)^2$$ over repeated training sets

$$\mathbb{E}(Y - \hat{f}_\mathcal{D}(x))^2 = \sigma^2 + \mathrm{Bias}^2 + \mathrm{Var}$$ add/subtract f, $\bar{f}$; cross terms vanish

$$\text{kNN: Bias} = \frac{1}{k} \sum_{j \leq k} f(x_{(j)}) - f(x)$$ Var $= \sigma^2/k$

Double descent: test error peaks at the interpolation threshold (params $\approx N$), falls again beyond.

Regularisation

$\min \widehat{R}(\theta) + \lambda\Omega(\theta) \equiv \min \widehat{R}$ s.t. $\Omega \leq c$ $\lambda \uparrow \Leftrightarrow c \downarrow$

$\Omega_2 = \|\theta\|_2^2$, $\Omega_1 = \|\theta\|_1$ ridge, lasso

$$\hat{w}_\lambda = \frac{\sum xy}{\sum x^2 + N\lambda}$$ 1-D ridge: shrinks, never 0

$$\hat{w} = \mathrm{sign}(z) \max(|z| - \lambda, 0)$$ lasso soft-threshold: exact zeros (diamond corners)

$$t^* = \arg\min_t \widehat{R}_{val}(\hat{\theta}_t)$$ early stopping; patience p = epochs without improvement

Subsets: 2^d; forward/backward $\approx d(d+1)/2$ fits; filters ignore interactions (XOR).

Selection optimism

$$\mathbb{E} \min_h \widehat{R}_{val}(\hat{f}_h) \leq \min_h R(\hat{f}_h)$$ winner's curse

$$\min(a,b) = \frac{a + b - |a-b|}{2}, \quad \mathbb{E}|\varepsilon_1 - \varepsilon_2| = \frac{2\sigma}{\sqrt{\pi}}$$

$$\Rightarrow \mathbb{E}\min(\widehat{R}_1, \widehat{R}_2) = R - \frac{\sigma}{\sqrt{\pi}}$$ two equal models; n models $\approx R - \sigma\sqrt{2\ln n}$

Metrics (positive = attack)

$$\mathrm{Acc} = \frac{TP+TN}{N}, \quad P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}$$

$$\mathrm{Spec} = \frac{TN}{TN+FP}, \quad \mathrm{FPR} = 1 - \mathrm{Spec}$$

$$\mathrm{BalAcc} = \frac{R + \mathrm{Spec}}{2}$$

$$F_1 = \frac{2PR}{P+R}, \quad \min(P,R) \leq F_1 \leq \frac{P+R}{2}$$

All-negative rule: Acc = 1− prevalence, $R = 0$.

Learning curves: both high and close = underfit; large gap shrinking with N = variance; still falling = data-limited.

4 Discriminative route

$$\sigma(s) = \frac{1}{1+e^{-s}}, \quad \sigma(-s) = 1 - \sigma(s)$$

$$\sigma'(s) = \sigma(s)\big(1 - \sigma(s)\big)$$

$$\log \frac{p}{1-p} = s_\theta(x) = w^\top x + b$$ log-odds is the score: linear boundary

$$\sigma(s) > t \iff s > \log \frac{t}{1-t}$$ costs shift the hyperplane, never bend it

$$\mathrm{softmax}(s)_c = \frac{e^{s_c}}{\sum_k e^{s_k}}$$ shift-invariant; C=2 gives $\sigma(s_1 - s_0)$

$$\log p_c = s_c - \log \sum_k e^{s_k}$$ log-softmax

$$\ell = -y \log \hat{p} - (1-y)\log(1-\hat{p})$$ binary CE

$$\frac{\partial \ell}{\partial s} = \hat{p} - y, \quad \frac{\partial^2 \ell}{\partial s^2} = \hat{p}(1-\hat{p})$$

$$\nabla_\theta \widehat{R}_{CE} = \frac{1}{N} \sum_i (\hat{p}^{(i)} - y^{(i)})\, x^{(i)}$$ convex; at the optimum $\sum_i (\hat{p}^{(i)} - y^{(i)}) = 0$

4 Generative route

$$p(c \mid x) = \frac{\pi_c\, p(x \mid c)}{\sum_k \pi_k\, p(x \mid k)}$$ Bayes' rule

$$= \mathrm{softmax}\big(\log \pi_c + \log p(x \mid c)\big)_c$$ generative score

$$\hat{\pi}_c = \frac{N_c}{N}, \ \hat{\mu}_c = \frac{1}{N_c} \sum_{y^{(i)}=c} x^{(i)}$$ MLE

$$\hat{\Sigma}_c = \frac{1}{N_c} \sum_{y^{(i)}=c} (x^{(i)} - \hat{\mu}_c)(x^{(i)} - \hat{\mu}_c)^\top$$

$$\mathcal{N}(x; \mu, \Sigma) = \frac{\exp[-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)]}{(2\pi)^{d/2}|\Sigma|^{1/2}}$$

QDA (own Σ_c**): quadratic boundary**

$$s_c = -\frac{1}{2}(x-\mu_c)^\top \Sigma_c^{-1}(x-\mu_c) - \frac{1}{2}\log|\Sigma_c| + \log \pi_c$$

LDA (shared Σ**):** $x^\top \Sigma^{-1} x$ **cancels, linear**

$$s_c = x^\top \Sigma^{-1} \mu_c - \frac{1}{2}\mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c$$

binary LDA: $w = \Sigma^{-1}(\mu_1 - \mu_0)$ boundary $w^\top x + b = 0$,

$$b = -\frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_0^\top \Sigma^{-1} \mu_0) + \log \frac{\pi_1}{\pi_0}$$

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-bc} \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$$

1-D, unequal variances: log-odds quadratic, up to two boundary points. Parameters: QDA $(C-1) + Cd + C\frac{d(d+1)}{2}$; LDA $(C-1) + Cd + \frac{d(d+1)}{2}$; Gaussian NB $(C-1) + 2Cd$; logistic $(C-1)(d+1)$.

Naive Bayes

$$p(x \mid c) = \prod_j p(x_j \mid c)$$ independence given the class

$$s_c = \log \pi_c + \sum_j \log p(x_j \mid c)$$ additive score, compute in logs

$$\frac{p(1 \mid x)}{p(0 \mid x)} = \frac{\pi_1}{\pi_0} \prod_j \frac{p(x_j \mid 1)}{p(x_j \mid 0)}$$ duplicated feature counted twice: overconfident, ranking kept

$$\hat{\theta}_{cj} = \frac{n_{cj} + \alpha}{\sum_\ell n_{c\ell} + \alpha V}$$ multinomial, Laplace

$$\hat{p}(x_j{=}v \mid c) = \frac{n_{cjv} + \alpha}{n_{cj} + \alpha K_j}$$ categorical; add an "other" level

$\hat{\sigma}_{cj}^2 \leftarrow \hat{\sigma}_{cj}^2 + \lambda s_j^2$ Gaussian NB variance smoothing

Prior shift and unfamiliar inputs

$\frac{r_d}{1-r_d} = \frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)} \cdot \frac{r_t}{1-r_t}$ needs calibrated r_t

$r_d = \frac{(\pi_d/\pi_t) r_t}{(\pi_d/\pi_t) r_t + \frac{1-\pi_d}{1-\pi_t}(1-r_t)}$ generative: swap π_c

$p(x) = \sum_c \pi_c \, p(x \mid c) < \tau \Rightarrow \text{reject}$ softmax confidence is only relative

Threshold for costs: same $\hat{p}$, other action. Reweighting/resampling: changes the estimate (tiny minority). Prior shift: rescale odds.

5 Averaging, trees, bagging, forests

$\hat{p}_{ens} = \frac{1}{M} \sum_m \hat{p}_m$ soft vote; hard vote counts labels

$\mathrm{Var}(\bar{e}) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{M}$ floor $\rho\sigma^2$; errors must differ

$T(x) = \sum_j \theta_j \mathbf{1}[x \in S_j]$ leaf: mean (sq.), median (abs.), majority (0–1)

$I_{\text{Gini}} = 1 - \sum_c \hat{p}_c^2$ 0 pure, max mixed

$I_{\text{ent}} = -\sum_c \hat{p}_c \log_2 \hat{p}_c$

$\Delta I = I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R)$ greedy split maximises it

$\mathbb{P}(i \notin \mathcal{D}^*) = (1 - \frac{1}{N})^N \to e^{-1} = 0.368$ 63.2% distinct per bootstrap

$\hat{f}_{OOB}(x^{(i)}) = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{f}_m(x^{(i)})$

$|\mathcal{M}_i| \approx 0.368M$; free validation

$\mathbb{P}(j \text{ offered at a split}) = \frac{m_{try}}{d}$

$\mathbb{P}(j \text{ absent at all } M \text{ roots}) = \left(1 - \frac{m_{try}}{d}\right)^M$

Forest = bagging + $m_{try} \approx \sqrt{d}$ candidates per split: $\rho \downarrow$ faster than $\sigma^2 \uparrow$. Tune m_{try}, depth on OOB/validation for the whole forest.

$\Delta_t = \frac{N_t}{N}\left[I(S_t) - \frac{N_L}{N_t} I(S_L) - \frac{N_R}{N_t} I(S_R)\right]$ MDI credit of split t

$\mathrm{MDI}_j = \frac{1}{M} \sum_m \sum_{t:\text{split on } j} \Delta_t$ training-based; favours many split points

$\mathrm{PI}_j = \mathrm{err}(\text{shuffle } x_j) - \mathrm{err}$ on OOB/val; correlated features mask; not causal

5 Boosting, XGBoost, stacking, MoE

$F_m = F_{m-1} + \eta T_m, \quad F_0 = \bar{y}$ sum of shrunken corrections

$r_m^{(i)} = -\frac{\partial \ell}{\partial F}\Big|_{F_{m-1}}$ squared loss: $r = y - F_{m-1}$; fit T_m to r

$\eta < 1$: each tree fixes part of the error; more rounds, smoother fit, better validation. η, M, depth are coupled; early-stop on validation (no OOB).

$\hat{p} = \sigma(F), \quad g = \frac{\partial \ell}{\partial F} = \hat{p} - y$

$h = \frac{\partial^2 \ell}{\partial F^2} = \hat{p}(1-\hat{p})$

$\ell(F + w) \approx \ell + gw + \frac{1}{2} h w^2$ Newton step per leaf

$w_q^* = -\frac{G_q}{H_q + \lambda} = \frac{\sum_{i \in q} (y^{(i)} - \hat{p}^{(i)})}{\sum_{i \in q} \hat{p}^{(i)}(1 - \hat{p}^{(i)}) + \lambda}$

$G_q = \sum_{i \in q} g^{(i)}$, $H_q = \sum_{i \in q} h^{(i)}$. Opposite-sign g in one leaf cancel: split them.

$\mathcal{L} = \sum_q \left[G_q w_q + \frac{1}{2}(H_q + \lambda) w_q^2\right] + \gamma Q$ XGBoost objective

$\phi_q^* = -\frac{1}{2} \frac{G_q^2}{H_q + \lambda}$ loss removed by leaf q

$\text{gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda}\right] - \gamma$ split iff > 0

$gw + \frac{1}{2} h w^2 = \frac{1}{2} h\big(-\frac{g}{h} - w\big)^2 + \mathrm{c}$ weighted LS on $-g/h$

multi-class: $g_k = \hat{p}_k - y_k, \; h_{kr} = \hat{p}_k(\delta_{kr} - \hat{p}_r)$ K trees/round

$\hat{p}_{stack} = \sigma\big(\beta_0 + \sum_m \beta_m z_m\big)$ z = out-of-fold predictions

$p(y \mid x) = \sum_m g_m(x) \, p_m(y \mid x)$ mixture of experts,

gate $g = \mathrm{softmax}(a(x))$; train on $-\sum_i \log$ of the mixture. Bagging: parallel, deep trees, cuts variance. Boosting: sequential, shallow trees, cuts bias. Neither fixes leakage, shift, or a missing feature.

Pairs he asks you to distinguish

Posterior vs decision: $\hat{p}$ is a number; the action uses $\hat{p}$ and the costs.

Leakage vs shortcut: leakage = unavailable at decision time or caused by the label (remove by construction); shortcut = legit feature correlated with y only in the collected data (break it with new data).

Training loss vs decision loss: CE estimates $\hat{p}$; 0–1/costs judge the action.

Cost threshold vs imbalance correction vs prior shift: same $\hat{p}$, new action / new estimate / rescaled odds.

Empirical vs population risk vs gap: observable / not / not.

Validation vs test: selects (optimistic once used) / touched once.

Under- vs overfitting: high train and val, small gap / low train, large gap.

Bias vs variance: error of the average prediction / spread across training sets.

Parameter vs hyperparameter: fitted on train / chosen on validation; depends on the procedure.

Discriminative vs generative: $p(y \mid x)$ directly, robust to wrong $p(x \mid y)$ / $p(x \mid y)p(y)$, data-efficient when right, handles missing x_j, flags unfamiliar x, swaps priors.

Hard vs soft vote: robust to one miscalibrated model / uses confidence.

Bagging vs boosting: variance, parallel, deep / bias, sequential, shallow.

MDI vs permutation: training impurity credit / validation reliance; neither causal.

Traps that cost points

Cost threshold is $C_{FP}/(C_{FP} + C_{FN})$, not $C_{FN}/(\cdot)$. kNN capacity falls as k rises. Scaler, imputer and feature selection fitted on all rows = leakage. Row-wise split with duplicated episodes = test in train. Best of n on the test set = validation score. Accuracy at 1% prevalence: all-negative scores 0.99. Resampled 50/50 training $\Rightarrow$ correct the prior before thresholding. Naive Bayes duplicates: odds squared. Softmax near 1 far from data $\neq$ familiar input. Forest floor $\rho\sigma^2$: more trees cannot pass it. Boosting has no OOB.